INCH-POUND

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19 December 1989
SUPERSEDING
MIL-I-27710C
18 AUGUST 1976

MILITARY SPECIFICATION

INDICATOR, ATTITUDE, REMOTE

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirments for remote attitude indicators.

1.2 Classification. Indicators shall be of the following types, as specified (see 6.2):

ARU-13/A	Red lighted
ARU-20/A	White lighted with improved coverglass coating.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEOR, Tinker AFB, OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6610

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2.1.1 Specifications and Standards. The following specifications and standards form a part of this document to extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS.

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-E-5400	Electronic Equipment, Airborne, General Specification For
MIL-C-5541	Chemical Conversion Coatings On Aluminum And Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-C-14806	Coating, Reflection Reducing, For Instrument Cover Glasses And Lighting Wedges
MIL-I-25467	Lighting, Integral, Aircraft Instrument, General Specification For
MIL-I-27160	Lighting, Instrument, Integral, White, General Specification For
MIL-C-83488	Coating, Aluminum, Ion Vapor Deposited

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, And Packing Levels
FED-STD-595	Colors

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-454	Standard General Reqirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirments For Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of

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MIL-STD-704	Electric Power, Aircraft Characteristics And Utilization Of
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-785	Reliability Program For Systems And Equipment Development And Production
MIL-STD-810C	Environmental Test Methods
MIL-STD-838	Lubrication Of Military Equipment
MIL-STD-882	System Safety Program For Systems And Associated Subsystems And Equipment: Requirements For
MIL-STD-889	Dissimilar Metals
MIL-STD-970	Standards And Specifications, Order Of Preference For The Selection Of
MIL-STD-2073-1	DoD Material Procedures For Development And Application Of Packaging Requirements
MS33556	Case, Aircraft Instrument, 3-1/4 Inch Size, Standard Dimensions for
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form Of

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

(Copies of MIL-STD-810C is available from Director, Naval Publications & Printing Service Office, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, Pa 19111.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualifications. The indicators furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.4 and 6.4).

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3.2 Components. The indicator shall consist of servo mechanisms for actuating the pitch and roll gimbals, an indicator display consisting basically of a fixed miniature airplane, bank angle dial, bank index and a moving two-colored spheroid background with a horizontal line dividing the two colors, a pitch trim knob and mechanism, external zeroing adjustments for pitch and roll, and a power-failure indication device.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-970.

3.4 Materials.

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. When used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the contracting activity. However if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.4.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.4.3 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt fog or atmospheric conditions likely to be met in storage or normal service.

3.4.3.1 Dissimilar metals. Unless protected against electrolytic corrosion by means of a protective coating or hermetic sealing, intimate contact of dissimilar metals shall be avoided. Dissimilar metals are defined in MIL-STD-889.

3.4.4 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and envirnomental conditions shall be avoided.

3.4.5 Corrosive fumes. The materials as installed in the indicator and under the service conditions specified herein shall not liberate deleterious fumes.

3.4.6 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encourged to the maximum extent possible without jeopardizing the intended end use of the item.

3.5 Design and construction. The indicator shall be so designed and constructed that the complete mechanism can be removed from the case as a unit, and adjustments and repairs can be easily made by the personnel of operating units and overhaul bases. It shall be built to withstand the strains, jars, vibrations and other conditions incident to shipping, storage, installation, and service.

3.5.1 Reliability program. The contractor shall establish a reliability assurance program and submit reports thereon in accordance with MIL-STD-785.

3.5.1.1 Reliability. The indicator shall have a minimum acceptable mean-time-between failures (MTBF) of at least 750 hours based on discrimination ratio of 2.0 and a confidence factor of 90 percent.

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3.5.2 Longevity. The indicator shall have a minimum operating life of 2,000 hours before wearout failures occur or the equipment consistently fails to meet the specified MTBF requirements.

3.5.3 System safety. The indicator shall be designed to comply with system safety criteria, considerations, and order of precedence specified in MIL-STD-882.

3.5.3.1 Failure modes and effect analysis. The functional relationships of components and interface with other equipments shall be analyzed to determine:

a. All failure modes which could result in Category III of IV hazards specified in MIL-STD-882.

b. Those modes of failure most likely to occur under expected service conditions including the effects of storage, shelf life, transportation and packaging as well as operational use.

c. Effects of failure modes identified above upon safety and performance of the indicator and of interfacing equipment.

3.5.3.2 Safety hazards. If hazards cannot be eliminated, alternative controls such as recommended changes to interfacing equipment shall be presented to the contracting activity for resolution. Design changes must be made to the indicator to eliminate, where possible, any safety hazards.

3.5.3.3 Fail-safe design. Failure of the assemble shall not cause degradation or failure of interfacing equipments.

3.6 Performance. The indicator shall be capable of meeting the requirements specified herein when subjected to the following conditions:

a. Operating temperatures ranging from -54° to $+71^{\circ}$C (-65° to $+160^{\circ}$F) and storage temperatures ranging from -62° to $+71^{\circ}$C (-80° to $+160^{\circ}$F)

b. Pressures ranging from 57 inches Hg down to 0.31 inch Hg, approximately 100,000-foot altitude

c. Exposure to salt sea atmosphere for a period of 50 hours

d. Vibration incident to service use

e. Fungus growth as encountered in tropical climates

f. Acceleration forces as anticipated

g. Immersion in water with pressure applied, or leak test conducted with an approved helium leak detector, without evidence of leakage exceeding that specified in 3.13.4

h. 500V dc for 10 seconds when applied as specified in 4.6.19.

3.6.1 Electromagnetic interference. The indicator shall be capable of meeting the interference requirements specified in MIL-STD-461 for Class 1B equipment.

3.6.2 Power Consumption. Power consumption shall not exceed 20 va.

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3.6.3 Magnetic effect. The indicator shall not cause a compass deflection in excess of 5° when tested as specified in 4.6.17.

3.7 Servomechanisms. The indicator shall be provided with two servomechanisms for actuating the pitch and roll gimbals.

3.8 Indicator display. The indicator display shall be in accordance with figure 1 and shall consist basically of a fixed miniature airplane, bank angle dial, bank index, and a moving, two-colored spheroid background with a distinct horizon line dividing the two colors. Unless otherwise specified, all colors shall be in accordance with FED-STD-595. The paint shall be such that the light will be sufficiently diffused to avoid highlight when the paint is illuminated with external or internal light sources and markings will be clear and sharp when viewed under such conditions. There shall be an expanded scale of pitch indications in the ratio of 1.5 to 1. The pitch angle specified herein refers to the indicatd angle and not the mechanical angle of the displacement.

3.8.1 Miniature airplane. The miniature airplane shall be in accordance with figure 2 or figure 3, as applicable. The rear edge shall be within 0.062 inch of the spheriod surface, and the wings shall follow the curvature of the spheriod. The top edge of the wings and the center dot shall be tilted back approximately 15° to 45° to facilitate meeting the lighting requirements. The model airplane shall be attached to the internal instrument structure by struts as shown on figure 2 or figure 3, as appicable. The attaching struts shall be black, color No. 37038. The miniature airplane shall have a black border of uniform width within 0.010 inch around the wings, center dot, and the painted portion of the "W". The white portion of the miniature airplane shall be painted white, color No. 37875.

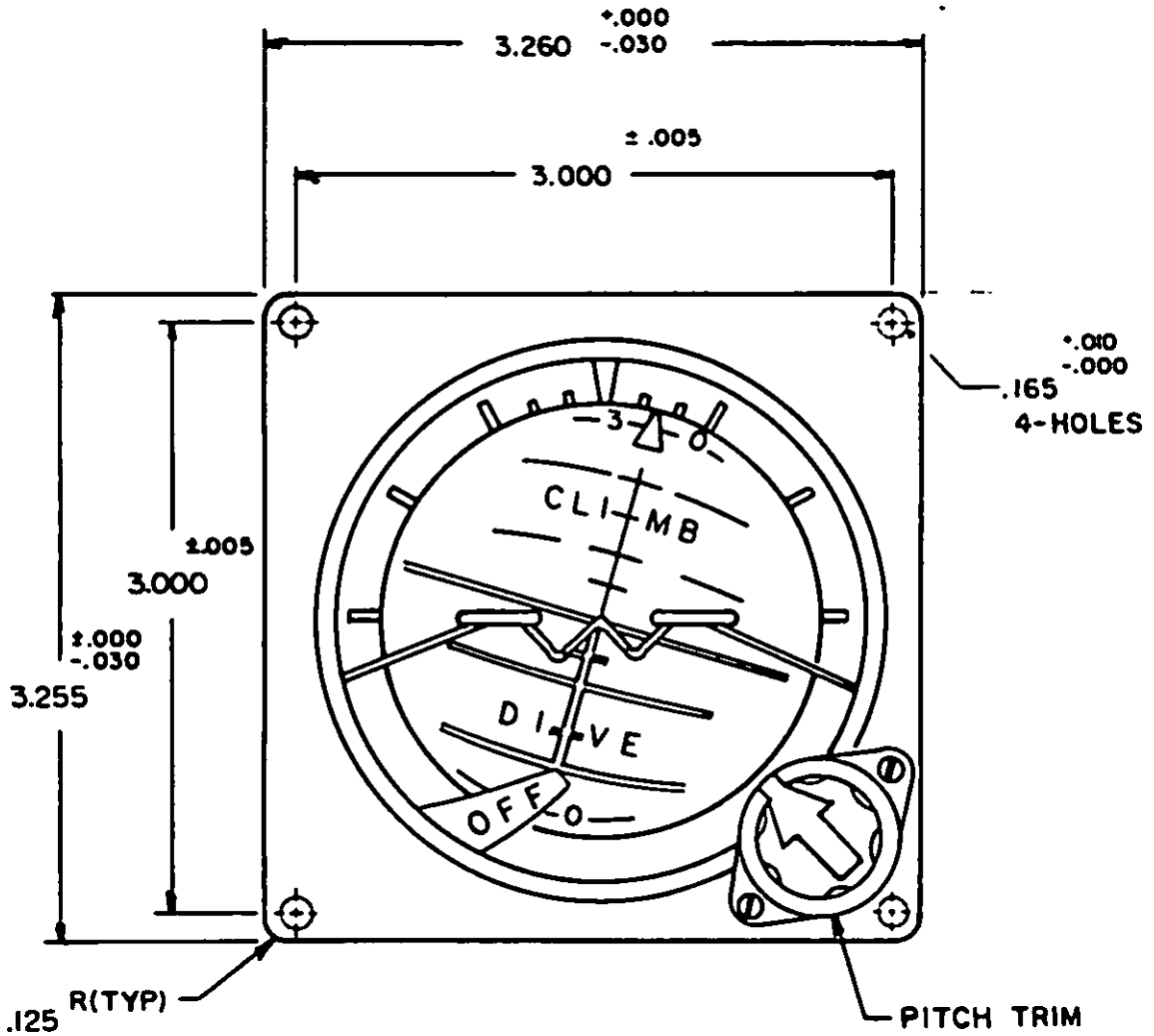
3.8.2 Bank angle dial. The bank angle dial shall be in accordance with figure 2 or figure 3, as applicable. The scale markings of the dial shall be color No. 37875.

3.8.3 Bank index. The bank index shall be in accordance with figure 2 or figure 3, as applicable. It shall be affixed to and rotate with the roll gimbal but shall not move with pitch deviations. The front surface of the bank index shall be in a plane parallel to the coverglass. This plane shall be in the same plane as the front surface of the small diameter of the bank dial with 0.047 inch. The arch described by the foremost portion of the bank index shall not deviate more than 0.047 inch radially from the calibrated small diameter portion of the bank scale. At zero roll indication, the bank index shall be aligned with and radially within 0.047 inch of the zero roll marking on the bank scale. The index shall be color No. 37875.

3.8.4 Mechanism mask. A mask with the visible surface shaped as the surface of the frustrum of a cone shall be attached to the roll gimbal. The mask shall extend from the bank angle dial and shall be approximately 2.375 inches on the large diameter. This edge of the mask shall be hidden by the inner diameter of the bank angle dial. The smaller diameter of the mask shall be closest to the spheriod and shall intersect the spheriod in such a manner as to provide a lateral opening of at least 2.187 inches. This edge of the mask shall conform to the shape of the spheriod to within 0.062 $\pm$0.031 inch.

3.8.5 Spheriod. The spheriod shall be mounted in the roll gimbal and shall rotate with the gimbal about the roll axis for roll indications and in relation to the roll gimbal about the pitch axis for pitch indications. The roll axis shall be perpendicular to the face of the indicator and the pitch axis shall be horizontal when the indicator is level and indicating a zero roll attitude. The horizon line on the spheriod shall be fixed to the spheriod at the zero pitch position. The

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DIMENSIONS IN DECIMAL

FIGURE 1. Indicator display.

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spheriod shall be driven in pitch in such a way that for a dive maneuver, the horizon line will go above the miniature airplane. The spheriod shall be in acordance with figure 4. The surface of the spheriod shall move 0.035 (nominal) inch for each degree of pitch change. The colors and markings shall be in accordance with figure 4 and the following requirements. Illumination may be accomplished by lighting within the spheriod. The distance from the outside surface of the coverglass to the spheriod shall not exceed 0.65 inch.

3.8.5.1 Colors and markings. Above the horizon line, the spheroid shall be light gray, color No. 36373 and all markings shall be color No. 37038. The horizon line shall be color No. 37875. Below the horizon line, the spheriod shall be color No. 37038 and all markings shall be color No. 36440. The lengths of the major pitch lines shall be as follows:

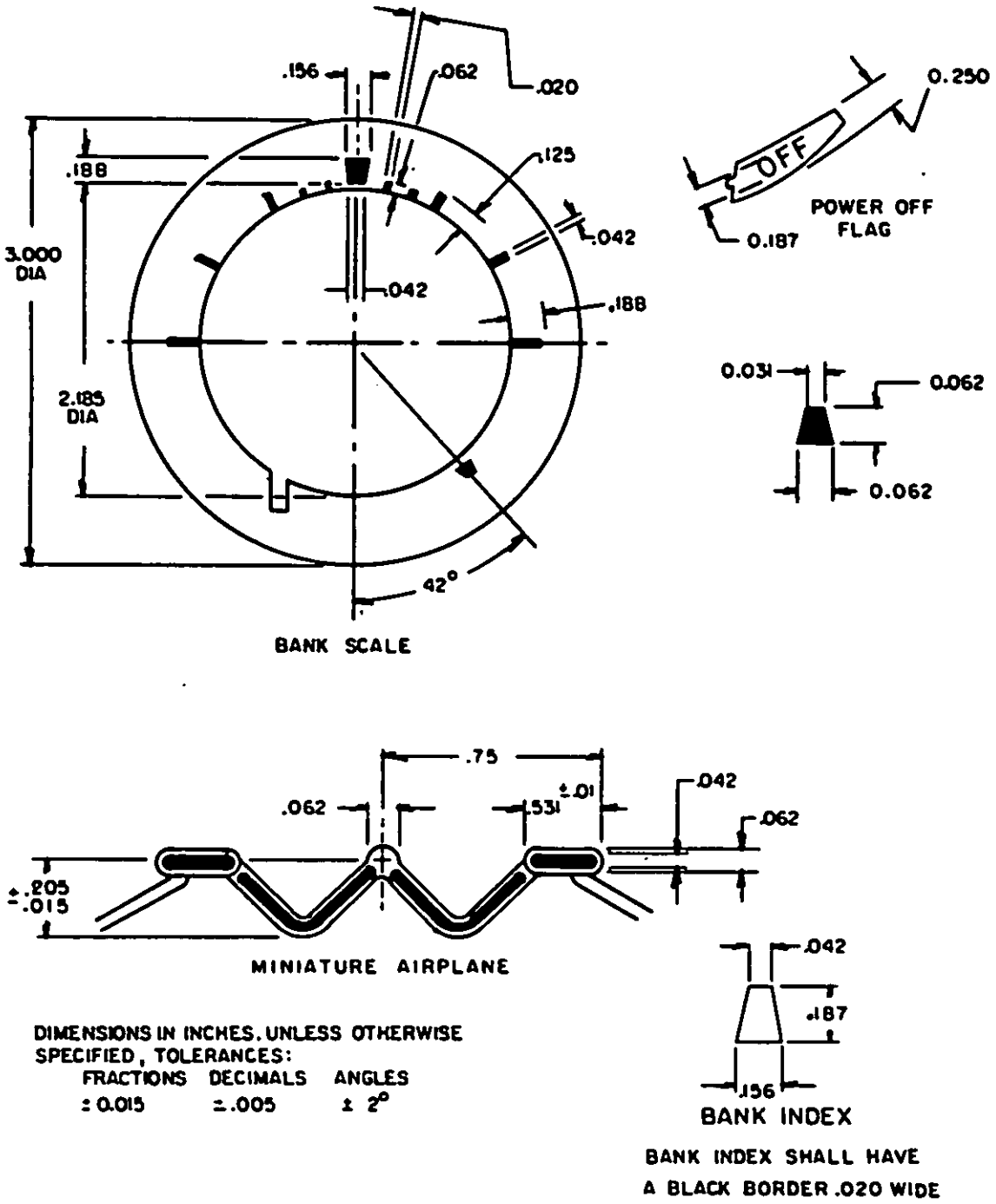
0°	(Horizon line).....Width of sphere
10°	.....1.687 inch
20°	.....1.531 inch
30°	.....1.343 inch
40°	.....1.156 inch
50°	.....0.969 inch
60°	.....0.781 inch
70°	.....0.594 inch
80°	.....Circle
85°	.....Solid dot

The center of the solid dots shall be the $\pm 90^{\circ}$ pitch position and from center to center shall represent 180° of pitch indication. The pitch indication shall be linear over the full 180° range which shall be spaced over 270° of the spheriod surface.

3.9 Pitch trim knob. A pitch trim knob shall be provided for adjusting the position of the spheriod. The knob shall be mounted on the lower right-hand side of the indicator and shall be marked as shown on figure 1. The knob shall be color No. 36231. The reference mark on the knob shall be color No. 37038, shall extend across the end of the knob, and shall continue along the side to a point on the knob that is adjacent to the index on the case at the zero pitch trim position. The index on the ARU-13/A indicator shall be a white trapeziod mark, color No. 37875 and the index on the ARU-20/A indicator shall be a yellow circle, color no. 33538. The knob shall have approximately 0.500 turn of freedom in the clockwise (cw) direction and 0.250 turn of freedom in the counterclockwise (ccw) direction from the zero pitch trim position. The freedom in each direction shall be limited by a positive stop. When the knob is rotated to the stop in the cw direction, the spheriod shall rotate to deflect the horizon line upward to indicate a dive of between 10° and 20°. When the knob is rotated to the stop in the ccw direction, the spheriod shall rotate to deflect the horizon line downward to indicate a climb of between 5° and 10°. The servo system shall operate properly for any pitch trim adjustment. The knob shall have sufficient friction to remain fixed when set but shall move smoothly without evidence of binding or erratic motion. The pitch trim circuit shall be so designed that if three indicators are connected to a single transmitter (pitch and roll), varying the pitch trim on any one of the indicators will not affect the other indicators by more than 0.250° in the pitch and roll axes. A minimum of 2 and a maximum of 6 ounce-inches of torque shall be required to rotate the pitch trim knob through the full range of the settable indications in both cw and ccw directions.

3.10 Zeroing adjustments. Means shall be incorporated to externally adjust the position of the spheriod $\pm 3^{\circ}$ minimum in pitch and roll. The intial setting of the pitch and roll indication at the time of delivery of the indicator shall indiate

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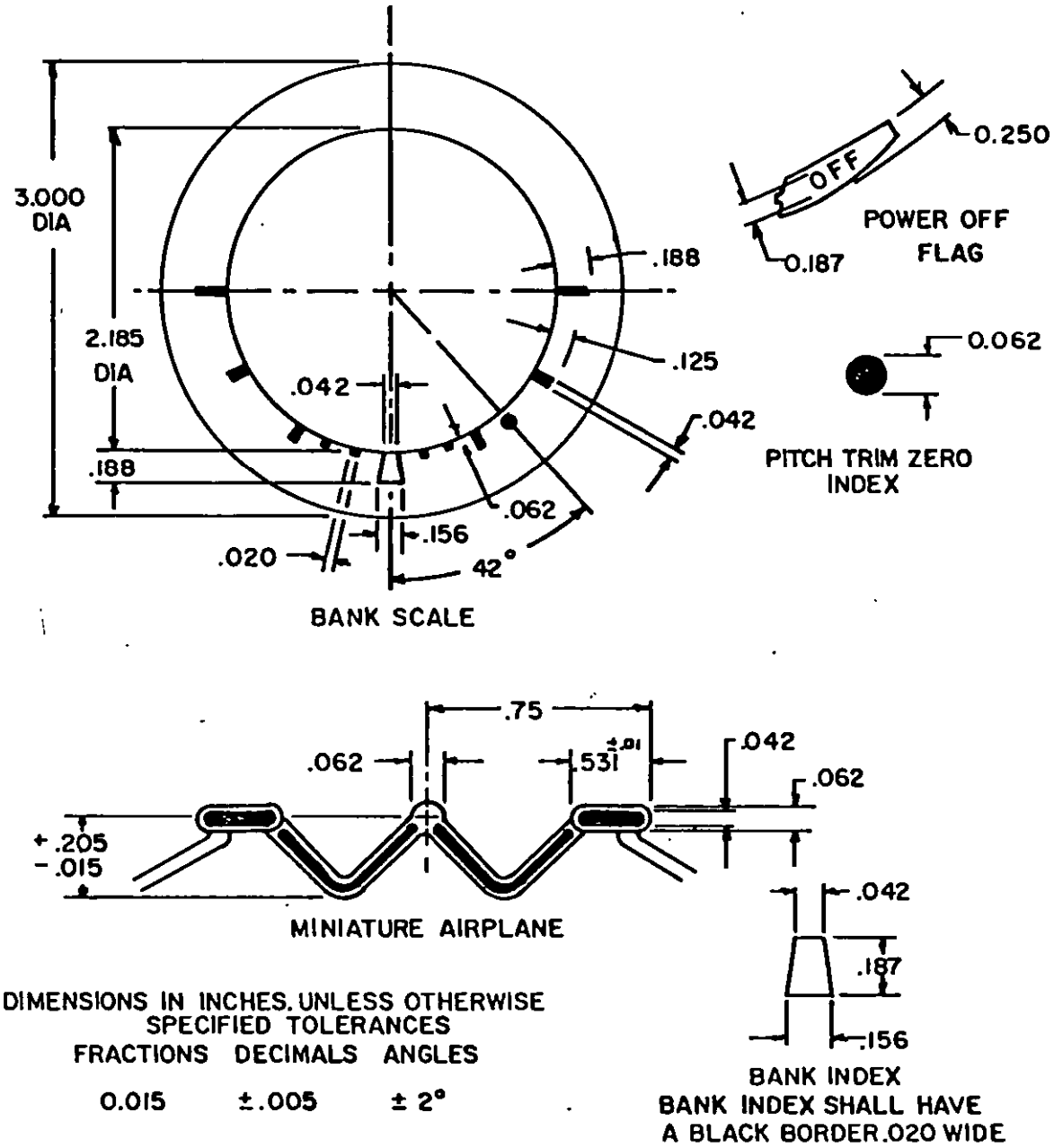


DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES:

FRACTIONS	DECIMALS	ANGLES
± 0.015	±.005	± 2°

FIGURE 2. Dial details for ARU-13/A indicator.

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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED TOLERANCES

FRACTIONS	DECIMALS	ANGLES
0.015	±.005	± 2°

FIGURE 3. Dial details for ARU-20/A indicator.

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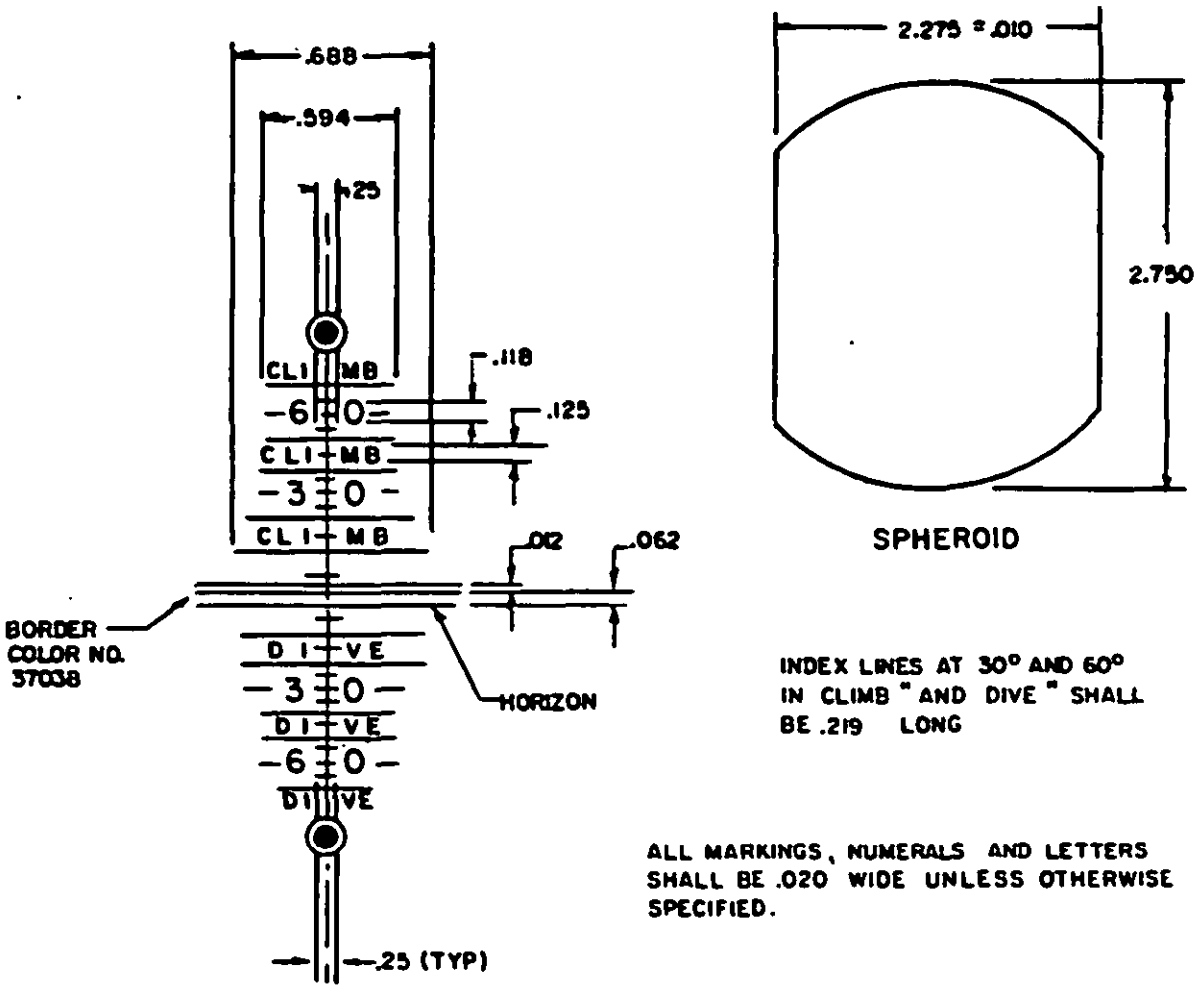


FIGURE 4. Spheroid dimensions and markings.

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zero pitch and roll with 0.500° when the indicator is connected to a test transmitter set at electrical zero. The pitch adjustment shall be combined with the pitch trim circuit by means of a settable knob. The roll adjustment shall be on the rear of the case and shall be adequately marked. It shall be equipped with stops and a locknut to secure the adjustment setting.

3.11 Power-failure indicator. A power-failure indicator shall be incorporated in the lower left of the indicator face on the bank angle dial as shown on figure 2 or figure 3, as applicable. The word OFF shall be in black letters, color No. 37038, on a fluorescent red background, color No. 116-1548 (Switzer Brothers, Cleveland, Ohio), or equal. A lead shall be brought out through the electrical connector through pin J. The circuit shall be such that when pin J is connected to pin A and power is applied, the warning will disappear. When the connection is broken or power is removed, the warning shall appear. The red background shall be illuminated under night lighting conditions. The lighting shall not be entirely dependent upon one light source.

3.11.1 Compatibility. The warning flag shall operate satisfactorily when connected in parallel with one to three indicators having the warning flag circuitry shown on figure 5.

3.12 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part number and changes thereto.

3.13 Indicator case. The indicator case shall be hermetically sealed and shall be in accordance with MS33556, except as shown on figures 1 and 6. The case shall be of sufficient strength to withstand an absolute pressure of 28 psi at temperature of 25°C (77° F).

3.13.1 Hermetic sealing. Hermetic sealing is the process by which the indicator is totally enclosed within a suitable metal structure or case and sealed airtight by fusing metals by welding, brazing, or soldering; fusion of ceramic materials under heat or pressure; or fusion of ceramic materials into a metallic support. Elastomeric or resinous materials or combinations thereof may be used for the coverglass seal provided the leak rate requirements are met under all conditions specified herein. Hermetic sealing shall not be accomplished by gaskets.

3.13.2 Coverglass. The coverglass shall be flat and free from discolorations, scratches, and striae that would interfere with reading the indicator. The thickness of the coverglass shall not exceed 0.200 inch.

3.13.2.1 ARU-20/A indicator. All visible glass surfaces of the ARU-20/A indicator shall be provided with a reflecton reducing coating that meets the requirements of MIL-C-14806.

3.13.3 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter, dewpoint -65°C (-85°F), at the filling pressure. The filling medium shall be 100 percent helium and 5 percent carbon dioxide. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere or 0.500 atmosphere, as required. Where practicable, the 100 percent helium filling medium shall be utilized.

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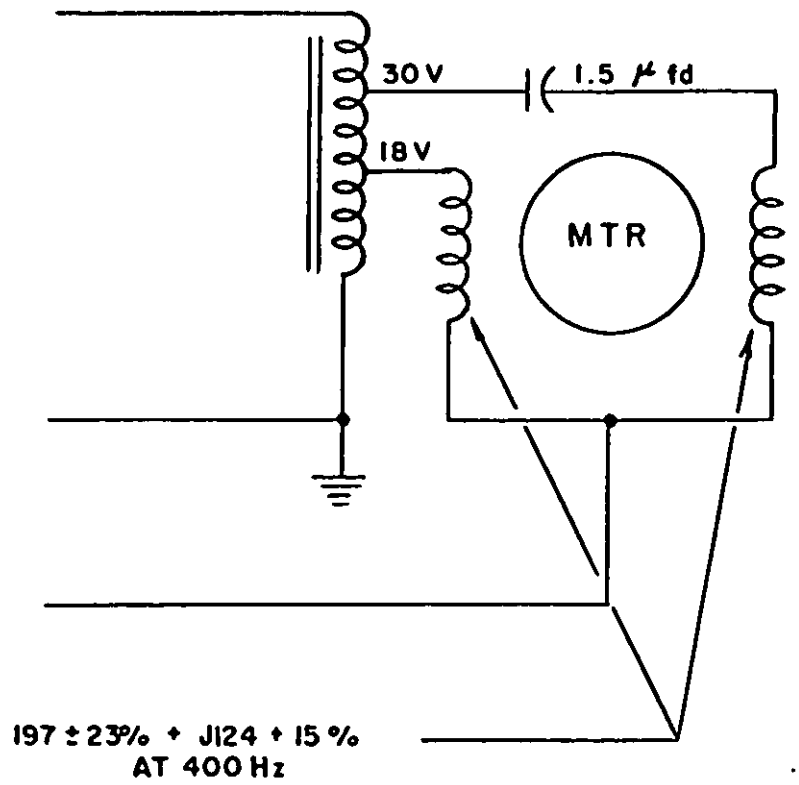


FIGURE 5. Warning flag circuitry.

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3.13.4 Leak rate. The case shall be so sealed that the maximum leak rate will not permit more than 10 percent contamination of the filling medium after 1,000 hours at a pressure differential equivalent to 1 atmosphere.

3.14 Lighting. Intergral lighting shall be provided in accordance with MIL-L-25467 or MIL-L-27160, as applicable. Supplemental lighting for the spheriod may be obtained by transillumination from within the spheroid. All production indicators shall be checked for proper lighting in accordance with 4.6.13.2. However, the gray portion of the spheriod shall at no time exceed 0.60 of the minimum brightness of the fixed white airplane.

3.14.1 External lighting. The indicator display shall be so designed that the display can be lighted by external means. This lighting capability shall be determined by use of Grime's fixture part No. B-4790, or equivalent.

3.14.2 Pins. Power for the lighting shall be in accordance with MIL-L-25467 or MIL-L-27160, as applicable. Pins K and L shall be utilized for lighting purposes. The lighting leads, or lead, shall not be grounded to the indicator case.

3.14.3 Prime Standard. Using parts that are identical to those used in production indicators, except parts that do not affect the lighting design, a lighted model of the indicator shall be constructed which shall meet all display requirements of MIL-L-25467 for the ARU-13/A indicator and MIL-L-27160 for the ARU-20/A indicator. The indicator model and recorded lighting data shall be submitted to the contracting activity for approval and if approved, the indicator model shall be certified as a prime standard.

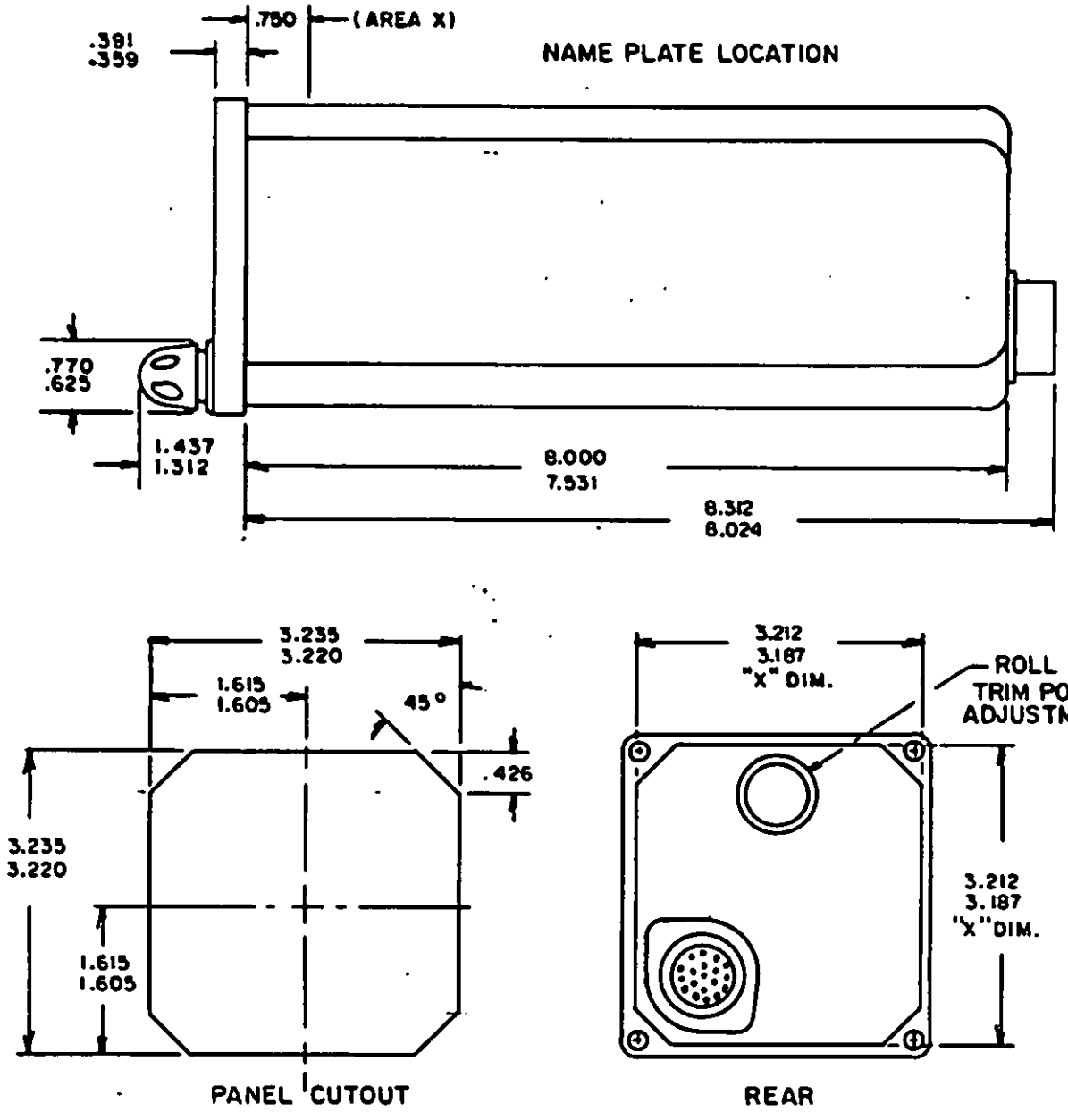
3.14.3.1 Prime standard life. A minimum of 12 average areas covering all representative areas of the face of the prime standard shall be measured for brightness and color prior to submission and at least every 50 hours of operation thereafter. After every 50 hours of operaton or after 120 days, new data shall be submitted to the contracting activity for approval.

3.15 Attitude actuation. The spheroid shall be rotated about the pitch and roll gimbal axes by means of geared servo motors until a null is attained between the control transformer synchros on the indicator axes and the transmitter synchros in the pitch and roll displacement reference. The freedom of the spheriod rotation shall be unlimited about the roll axis and a minumum of $\pm 90^{\circ}$ of indication about the pitch axis. Stops shall be incorporated in pitch so that the indication will not become unsynchronized with the pitch input signal when continuous rotation of the pitch input occurs. This continuous input will exceed the $\pm 90^{\circ}$ pitch range of the indicator and again come within range due to continuous rotation. The indicator shall not be damaged due to continuous pitch input rates of 90° per second in either direction.

3.15.1 Synchro control transformer. The synchro control transformers on the pitch and roll axes shall be high-impedance synchros. When one to four indicators are connected to a single test transmitterr synchro per axis, no change in performance shall be noted.

3.15.2 Standard test transmitter. The standard test transmitter shall be an Eclipse Pioneer type AY200-1, or equal, high-precision synchro with minimum and known errors. The rotor of the snychro shall be settable and rotatable through 360°. These settings shall be readable to within 6 minutes of arc and each degree of scale marking shall represent 1° of synchro rotor rotation. The following procedure shall be used to calibrate the test transmitter:

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CONNECTOR, BENDIX CORP.
PART NO. PT02H-14-19P, OR
EQUAL

UNLESS OTHERWISE SPECIFIED,
DIMENSIONS IN INCHES
TOLERANCES: DECIMALS ± .005

FIGURE 6. Indicator case.

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a. The rotor leads shall be designated as H and C and one stator lead designated as X and connected to C. When 26V is applied across H and C (C being ground) the scale shall be positioned on zero when the voltage across H and X is maximum and the voltage across the remaining two stator leads (X and Y) is minimum. The voltage across X and H increases before it decreases for increasing numbers of the scale.

b. The rotors of the standard test transmitters shall be excited from an auto-transformer or electrical equivalent which in turn is excited by the power supply voltage across the indicator electrical connector pins A and B. Rotor lead C shall be common with pin A.

3.15.3 Pitch and roll sensing. With the standard test transmitter set at a scale reading of 180° and connected to the ARU-13/A indicator as shown in tables I and II, the pitch and roll indicatons shall be zero. Increasing scale settings of the standard test transmitter shall result in right bank and climb indications, while decreasing the scale settings shall result in left bank and dive indications.

TABLE I. Roll servo connections.

Test Transmitter Terminal	Indicator Terminal of Power Input Connector	Rotor Excitation of Standard Test Transmitter
Hot (H)		26V tap-off from single-phase power supply to pins A and B of indicator
Cold (C)		
X	D	
Y	E	
Z	F	

TABLE II. Pitch servo connections.

Test Transmitter Terminal	Indicator Terminal of Power Input Connector	Roto Excitation of Standard Test Transmitter
Hot (H)		26V tap-off from single-phase power supply to pins A and B of indicator
Cold (C)		
X	R	
Y	H	
Z	P	

3.16 Followup operation. The electrical and mechanical design of the ARU-20/A indicator shall be such that in the process of followup, the indications in the pitch and bank will be smooth without hunting, oscillating, or jumping more than 0 500° total amplitude in pitch and 1° total amplitude in roll when connected to a standard transmitter. The same electrical and mechanical design requirements shall apply to the ARU-13/A indicator except the hunting, oscillating, or jumping shall not exceed 0.125° total amplitude in pitch and roll. There shall be no sustained electrical or mechanical oscillatons in either axis greater than 0.250° total amplitude for any indications. The servo mechanism shall be 70 to 100 percent damped and shall not lag the signal source by more than 10° at the maximum rate of servo operation.

3.16.1 Followup rate. The rate at which the motors drive the gimbals shall not be less than 300° per second in roll nor 180° per second in pitch.

3.16.2 Followup accuracy. The pitch and roll indications shall follow the imput

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signals from their respective test transmitter synchro to within 1° for positions up ±30° in roll and pitch and 2° for all other positions.

3.17 Amplifier channels. The amplifier shall be an intergral part of the indicator. There shall be three transistor or magnetic amplifier stages. The output stage and at least one other stage shall be redundant. The indicator shall continue to operate with reduced performance should partial failure of the amplifier occur.

3.18 Electrical power requirements. The power supply to the indicator shall be 115V, single-phase, 400 Hz in accordance with MIL-STD-704. Power input pin connections shall be in accordance with table III.

3.18.1 Electrical power variation. The indicators shall operate properly and continuously at voltages ranging from 103V to 127V and at frequencies ranging from 380 to 420 Hz for the ARU-13/A and from 320 to 480 Hz for the ARU-20/A. The indicators shall operate at any combination of voltage and frequency specified herein and under the transient variations specified in MIL-STD-704.

3.19 Electronic Components. The electronic components and application thereof shall be in accordance with MIL-E-5400.

TABLE III. Power input connections.

Electrical Pins	Connections
A	Neutral (grounded to case)
B	115V, 400 Hz, single-phase power in phase with excitation to test transmitter synchros
C	No connection
(X) D (Y) E (Z) F	The three stator leads from the roll test transmitter synchro
(X) R (Y) H (Z) P	The three stator leads from the pitch test transmitter synchro
J	Power failure warning (connect to pin A to complete circuit)
K	Internal lighting power input
L	
M	Pitch null test terminal (pitch error voltage between this pin and pin A)
N	Roll null test terminal (roll error voltage between this pin and pin A)
G-S-T-U-V	No connection

3.20 Electrical Connector. The electrical connector shall be Bendix Scintilla P. T. type, or equal. The connector shall be affixed to the case as shown on figure 6,

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except that it may be in either of the lower corners of the case. If shall be hermetically sealed and shall mate with a Bendix Scintilla type PT06P-14-19S, or equal. The connector shall be equipped with a cap or plug an gasket in a manner that will protect the electrical surfaces from water vapor. The electrical connections shall be in accordance with table III.

3.21 Lubrication. The indicator shall be lubricated in accordance with MIL-STD-838.

3.22 Markings. Unless otherwise specified, the dimensions of all markings on the indicator shall be in accordance with figures 1 through 4 and all visible portions of the indicator shall conform to color No. 37038.

a. All markings shall be clear and sharp. Dimensions shall be as specified and the width of any given mark shall not vary noticeably throughout the length of the mark unless otherwise specified. Markings which for some given indication become adjacent and are specified to have the same basic dimension at the adjacent point shall not be noticeably different at that point.

3.23 Numerals and letter. Unless otherwise specified, all numerals and letters on the indicator, other than the nameplate, shall be in accordance with MS33558.

3.24 Soldering. Soldering shall be accomplished in accordance with MIL-STD-454.

3.25 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.26 Weight. The weight of the indicator shall not exceed 3.75 pounds.

3.27 Finishes and protective coatings.

3.27.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

a. Dials, small holes, and case inserts need not be anodized.

b. Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

c. Where the primary purpose of the treatments is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 in lieu of anodizing.

d. Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

e. When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

f. Parts in a hermetically sealed container need not be anodized.

g. When the part is plated with tin over a copper flash, the part need not be anodized.

3.27.2 Steel parts. Steel parts shall be coated with Ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488. Type I or II as applicable and

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of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

3.28 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.29 Workmanship. The indicator, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner in accordance with MIL-STD-454 requirement 9. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts, and tightness of assembly screws and bolts.

3.29.1 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactority headed.

3.29.2 Cleaning. The indicator shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

3.29.3 Dimensions. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.29.4 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertian conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are

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classified as follows:

a. Qualification inspection (see 4.4).

b. Qualification conformance inspection (4.5).

4.3 Inspection Conditions: Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in paragraph 4.3.1 in this specification.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature, approximately 25°C (77°F). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Voltage input. Unless otherwise specified, the indicator shall be tested with 115V 400 Hz (normal) single-phase power.

4.3.3 Properly connected. When the indicator is referred to as being properly connected, it shall be understood that it is connected to a test fixture that is providing a lighting circuit voltage variable from 5V (ac or dc) to approximately zero volts, synchro signals are in accordance with 3.15.1, 3.15.2, and 3.15.3, a-c power is as required, and pin J is jumped to pin A through a closed switch.

4.3.4 Visual observation. Unless otherwise specified, all visual observations of indicator performance shall be made at a distance of 2 feet.

4.3.5 Burn-in. Prior to initiation of production testing, the contractor shall operate the indicator for at least 10 hours. Any failures that occur during this time shall be considered irrelevant. The failures and test time shall be recorded and reported.

4.4 Qualification inspection.

4.4.1 Qualification inspection samples. The qualification inspection samples shall consist of three indicators. Samples shall be identified with the manufacturer's own part and any additional information required by the letter of authorization.

4.4.2 Qualification inspections. The qualification inspections shall consist of all tests specified under 4.6.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Individual test .....See 4.5.1.

b. Sampling plans and tests.....See 4.5.2.

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product.....See 4.6.1

b. Starting.....See 4.6.2

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c. Pitch and bank zero.....See 4.6.3

d. Pitch trim.....See 4.6.4

e. Zeroing adjustments.....See 4.6.5

f. Sensitivity.....See 4.6.6

g. Followup rate.....See 4.6.7

h. Followup accuracy.....See 4.6.8

i. Gimbal freedom.....See 4.6.9

j. Leak rate.....See 4.6.10

k. Power warning indicator.....See 4.6.11

l. Fogging.....See 4.6.12

m. Lighting.....See 4.6.13

4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A. One indicator shall be selected at random from each 100 or less items produced on the contractor or order. The selected indicator shall be subjected to the following tests as described under 4.6:

a. Individual tests.....See 4.5.1

b. Power consumption, weight, and size.....See 4.6.14

c. Power variation.....See 4.6.15

d. Followup operation (except damping).....See 4.6.16

e. Magnetic effect.....See 4.6.17

f. Acceleration (with power applied).....See 4.6.18

g. Dielectric strength.....See 4.6.19

h. Vibration error.....See 4.6.21

i. Horizontal line and bank index null position shift.....See 4.6.20

j. Low temperature operation.....See 4.6.22

k. High temperature operation.....See 4.6.23

4.5.2.2 Sampling plan B. Unless otherwise specified, 3 indicators shall be selected at random from the first items on the contractor or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests.....See 4.5.2.1

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b. High temperature exposure.....See 4.6.24

c. High altitude-low temperature.....See 4.6.25

d. Temperature shock.....See 4.6.26

e. Fungus.....See 4.6.27

f. Salt fog.....See 4.6.28

g. Vibration failure.....See 4.6.29

h. Acceleration (without power applied).....See 4.6.30

i. Pressurization.....See 4.6.31

j. Electromagnetic interference.....See 4.6.32

k. Damping.....See 4.6.16.3

l. Reliability and MTBF.....See 4.6.33

m. Longevity.....See 4.6.34

NOTE: At the option of the contracting activity, the tests may be divided into three groups as follows, to be conducted on three sets of three instruments each, if a need exists to expedite tests. Each indicator shall meet the requirements of all the individual tests after completing the group of specific tests to which submitted.

Group I	Group II	Group III
Sampling plan A tests		Vibration failure
Electromagnetic interferencee	Fungus	Acceleration (without power
High temperature exposure		applied)
High altitude-low temperature	Salt fog	Damping
Temperature shock	Pressurization	Reliability and MTBF
		Longevity

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the contracting activity of all defects likely to be found and methods of correcting them. The contractor shall be ultimately responsible for the correction of defective items already accepted. The government with the aid of the contractor will determine the most efficient method of correcting the items.

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4.6 Test methods

4.6.1 Examination of product. Prior to sealing, the indicator shall be inspected to determine compliance with 3.4, 3.12, 3.13.2, 3.13.3, 3.19, 3.20, 3.21, 3.22, 3.23, 3.24, 3.25, 3.27, 3.28, and 3.29. This inspection need only be performed once on each unit produced. Upon final assembly, the indicator shall be inspected to determine complaince with the requirements specified herein with respect to materials, workmanship, marking, weight and outline dimensions.

4.6.2 Starting. The indicator shall be properly connected and power applied. The warning flag shall disappear and the indicator shall be ready for immediate operation.

4.6.3 Pitch and bank zero. The standard test transmitters in the pitch and roll axes shall be set at the electrical zero position. Power shall be applied and the indicator shall follow-up to the zero indications in pitch and bank within $\pm 0.500^{\circ}$ from any random indication in either axis. When the indication is stable under these conditions, the error signal voltage from the two control transformers shall be as near as practicable to minimum voltage. This test shall be repeated and the indication shall return to within $\pm 0.250^{\circ}$ of its previous indication.

4.6.4 Pitch trim. The pitch trim knob shall be adjusted to its limits of 0.250 turn ccw and 0.500 turn cs and the deflection of the horizon line observed. The respective deflections shall be from 5° to 10° down and from 10° to 20° up from the level flight indications. With the pitch trim knob in the zero pitch trim position, the indicator shall indicate zero pitch within ± 0.500 of the horizon line width. The pitch trim torque shall be between 2 and 6 ounce-inches. Two indicators shall be connected to the same transmitter and the pitch trim knob on one end of the indicators moved through its range of motion. There shall be not more than 0.500° change in the indication of the other indicator.

4.6.5 Zeroing adjustments. The indicator shall be properly connected to the standard transmitters which shall be set at electrical zero. The roll zeroing adjustment shall be moved to its extremes and deflection of the bank index noted. The deflection shall be not less than $\pm 3^{\circ}$ from the zero position. The pitch test transmitter shall be rotated 3° in climb and then 3° in dive. In each case, the pitch trim knob shall be loosened on its shaft and the shaft rotated until the indication is zero. The pitch trim knob shall then be tightened to the shaft at the zero position and shall meet the requirements of 4.6.4, except the requirement that there shall be not more than 0.500° change in indication of the other indicator shall not apply. This test shall not be repeated when specified under sampling plan B except as part of the longevity test.

4.6.6 Sensitivity. The test transmitter for the roll axis shall be moved 0.250° in right bank and returned to the original position and 0.250° in the left roll and returned to the original position. The test transmitter for the pitch axis shall be moved 0.250° in dive and returned to the original position an 0.250° in climb and returned to the original position. There shall be a smooth perceptible movement of the indicator bank index and horizon line for each movement of the appropriate test transmitter. The test shall be repeated in dive and climb with the pitch trim in each of its extreme positions, except the test transmitter shall be moved 0.500° in each direction.

4.6.7 Followup rate. The indicator shall be properly connected to a remote signal source or standard test transmitter. The follow-up rates in pitch and roll shall be determined. The computed rate in the roll axis shall be not less than 300° per

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second and the computed rate in the pitch axis shall be not less than 180° per second.

4.6.8 Follow-up accuracy. The rotor of the standard test transmitter in the roll axis shall be displaced to five positions other than the zero position. The indication of the indicator shall position itself with the rotor of the standard test transmitter within ±1° for each position between 0° and 30° left and right bank and within 2.0° for any position over the remaining 300°. The rotor of the standard test transmitter in the pitch axis shall be displaced to five positions other than the zero position. The indication of the indicator shall position itself with the rotor of the standard test transmitter within ±1° for each position between 0° and 30° dive or climb, and within 2.0° for any position between 30° and 90°.

4.6.9 Gimbal freedom. The indicator shall be properly connected to the remote signal or standard test transmitter and the transmitter positioned 360° in roll and ±90° in pitch. The indicator shall follow smoothly without noticeably sticking, overshooting, or hunting. The pitch axis test transmitter signal shall be rotated through 360°. The pitch indication shall synchronize with the input signal when it is within a ±90° range of the pitch indication. The indicator shall not be damaged by the input signal exceeding the ±90° pitch indication range.

4.6.10 Leak rate. The instrument case shall be tested for leaks by means of a mass spectrometer type of helium detector or other suitable means approved by the contracting activity. The maximum leak rate shall not exceed that specified in 3.13.4.

4.6.11 Power warning indicator. The indicator shall be properly connected with no power applied. Power shall be applied and the flag shall disappear. The power shall be disconnected and the flag shall immediately appear. Power shall be reapplied and the jumper from pin J to A disconnected; the flag shall appear.

4.6.11.1 Compatibility. The indicators shall be connected in parallel with three other attitude indicators having a warning flag circuitry conforming to figure 5. When the lead which parallels all J pins to indicators is not connected to the aircraft ground, or pin A, the warning flag shall disappear. Conversely, when the lead that parallels the J pins is connected to pin A, the warning flag shall appear.

4.6.12 Fogging. The indicator shall be energized in a 71° ±2° C (160° ±4°F) controlled ambient for a minimum period of 1 hour. After this period and while at this temperature, an ice cube shall be rubbed on the indicator glass face for a period of from 30 ±5 seconds. The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence of water or oil fog. Evidence of fogging shall be cause for rejection.

4.6.13 Lighting

4.6.13.1 Prime standard an qualifications. ARU-13/A indicators used as prime standards and ARU-13/A indicators submitted for qualification shall be tested as specified in MIL-L-25467. ARU-20/A indicators used as prime standards and ARU-20/A indicators submitted for qualification shall be tested in accordance with MIL-L-27160.

4.6.13.2 Production testing. Lighting tests on production indicators shall be accomplished as follows. Particular attention shall be given to individual segments of the indicator when making the specified comparisons. Any indicator that does not meet the visual comparison tests shall not be approved unless the indicator is

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subjected to all brightness, color, and stray light tests listed in MIL-L-25467 for ARU-13/A indicators and MIL-L-27160 for ARU 20/A indicators and sufficient data has been recorded to substantiate conformance to the tests.

a. A voltage of 5V shall be applied to the prime standard and to the production indicator being tested. The two indicators shall be compared and shall visually appear identical in color.

b. A voltage of 2.7V shall be applied to the prime standard. The voltage applied to the indicator being checked shall be varied until it appears identical in brightness to the prime standard. The voltage of the indicator being checked shall be read and recorded and shall be between 2.60 and 2.80V.

4.6.14 Power consumption, weight, and size. Power consumption of the indicator shall be measured with suitable meters and shall not exceed 20 va. The power factor shall be maintained between 0.70 and 1.00. The weight of the indicator shall be measured and shall not exceed 3.75 pounds. The outline dimensions shall be measured and shall be in accordance with figures 1 and 6. All results shall be recorded.

4.6.15 Power variation. With power applied, the indicator shall be properly connected and subjected to the transient power condition specified in MIL-STD-704, Curve I at 100 milliseconds and also 5 seconds. There shall be no damage to the indicator. The indicator shall then be subjected to the test specified in 4.6.3, 4.6.6, 4.6.9, and 4.6.11 at voltages and frequencies of 103V 380 Hz, 127V 380 Hz, and 103V 420 Hz for the ARU-13/A indicator and at all combinations of 103 and 127V and 320 and 480 Hz for the ARU-20/A indicator. The tolerance for 4.6.3 shall be 0.750° for zero bank and for 4.6.6 shall be 1° in roll and 0.750° in pitch.

4.6.16 Follow-up operation. Suitable test equipment shall be connected to the appropriate test pins of the indicator and any hunting, jumping, lagging, or overshooting of the indication when the indicator is made to follow the standard transmitter at various constant rates in both the pitch and roll axes shall be observed and recorded. This shall be possible with the indicator sealed. Pen recorders shall not be used to determine follow-up rates and oscillation.

4.6.16.1 Indication lag. In the pitch axis, the indication shall not lag behind the pitch remote input transmitter by more than 5° for a shaft speed of 90° per second nor more than 10° for a shaft speed of 180° per second. In the roll axis, the indication shall not lag behind the roll remote input transmitter by more than 4° for a shaft speed of 90° per second nor more than 10° for a shaft speed of 300° per second.

4.6.16.2 Hunting and jumping. There shall be no perceptible hunting or jumping of the indicator bank index or horizon line for cw and ccw rates of 1.5° per second or zero input. In addition, the manufacturer shall conduct a servo-error survey over the full range up to $\pm 90^{\circ}$ per second in pitch and $\pm 240^{\circ}$ per second in roll. The hunting and jumping shall not exceed a peak-to-peak amplitude of 0.125° for input speeds up to $\pm 20^{\circ}$ per second, 0.500 for input speeds up to $\pm 90^{\circ}$ per second, 1° for input speeds up to $\pm 240^{\circ}$, The testing rates shall include the following rates in both directions:

Pitch 0°, 1.5°, 20°, 60° andd 90° per second.

Roll 0°, 1.5°, 20°, 60°, 90°, 120°, 180°, and 240° per second.

Each test at 1.5° per second shall start at an indicated $\pm 60^{\circ}$ and proceed to the

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opposite $\pm 60^{\circ}$ on the indicator. At zero roll and pitch inputs, the pitch trim shall be moved to both extremes and the hunting and jumping shall be less than 0.125°, peak-to-peak amplitude. In addition, a 1.5° per second rate shall be simultaneously applied to both the roll and the pitch inputs. All combinations of cw and ccw shall be tested and the peak-to-peak amplitude shall be less than 0.125°. A pitch input of 90° per second shall be applied for a minimum of 2 revolutions in both directions. There shall be no damage from this continuous input.

4.6.16.3 Damping. The damping of the indicator servos shall be determined and shall be 70 to 100 percent critical damping.

4.6.17 Magnetic effect. This test shall be made first with no power applied to the indicator and then with the indicator operating on rated power. The indicator shall be held in various positions magnetically east or west of and 5 inches from the center of a free magnet, approximately 1.500 inches long, in a magnetic field with a horizontal intensity of 0.18 $\pm$0.01 oersted. The maximum deflection of the magnet shall not exceed 5°. An aircraft compass with the compensating magnet removed may be used as the free magnet for this test.

4.6.18 Acceleration (with power applied). The indicator shall be properly connected with power applied and accelerated along its normal vertical axis with a force of 5g. The attitude display shall not vary by more than 1° in pitch or roll from the initial indication. This test shall be conducted with the attitude set at zero pitch and roll and with each axis displaced to an indicated 60°.

4.6.19 Dielectric strenght. A potential of 500V dc shall be applied between isolated pins and between pins and the case for a period of 10 seconds. There shall be no breakdown of insulation.

4.6.20 Horizon line and bank index null position shift. The indicator shall be properly connected to standard test transmitters in the pitch and roll axes and the tansmitters set to drive the indications in both axes to the zero position. The input voltage and frequency shall be varied within the following limits and combinations thereof:

Frequency	380 to 420 Hz
A-C voltage	103 to 127V

The horizon line shall not shift from null position more than $\pm$0.062 inch. The bank index shall not shift from normal null position more than 0.500 of the narrow end of the pointer width.

4.6.21 Vibration error. The indicator shall be connected to test transmitter and mounted in its normal operation position. While being operated, the indicator shall be vibrated with a circular motion in a plane inclined 45° to the horizontal plane at an amplitude between 0.003 and 0.005 inch, and the frequency shall be varied slowly from 5 to 50 Hz. No condition of applied vibration shall cause pointer oscillation or variation in excess of the values specified in 4.6.21.1 and 4.6.21.2, respectively.

4.6.21.1 Roll and pitch oscillation. The maximum total amplitude of roll and pitch indicated oscillation throughtout the frequency range shall be noted and shall not exceed 0.500°.

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4.6.21.2 Roll and pitch variation. During vibration, the maximum variation of the roll and pitch indication from the original indication before vibrating, throughout the frequency range, shall be noted and shall not exceed 0.500°.

4.6.22 Low temperature operation. The indicator shall be properly connected to the standard test tansmitter, with no power applied, and placed in a chamber at a temperature of -54° ±2°C (-65° ±4°F) for a period of 4 hours. At the end of this period and while the indicator is at the reduced temperature, power shall be applied for a period of 5 minutes. At the end of this period while still at the reduced termperature, the indicator shall be subjected to and shall meet the tests specified in 4.6.3, except the tolerance shall be ±1°, and 4.6.6, except the tolerances in pitch and roll with the pitch trim knob in the zero position shall be 0.500° and the tolerance in dive and climb with the pitch trim knob in each of its extreme positions shall be ±1°.

4.6.23 High temperature operation. The indicator shall be properly connected with power applied, placed in a chamber at a temperature of +71° ±2° (+160° ±4°F), and operated for a period of 4 hours. At the end of this period and while at the high temperature, the indicator shall meet the tests specified in 4.6.3, except the tolerance shall be ±1°, and 4.6.6, except the tolerance at the zero pitch trim shall be 0.500°.

4.6.24 High temperature exposure. The indicator shall be properly connected but no power applied and subjected to a high temperature exposure test in accordance with MIL-STD-810C, Method 501.1, procedure 1, except that in step 3, the temperature shall be maintained for 24 hours and step 7 shall not apply. At the end of the 24-hour period and while at the high temperature, the indicator shall meet the test specified in 4.6.9.

4.6.25 High altitude-low temperature. The indicator shall be subjected to a high altitude-low temperature test. The temperature chamber shall be reduced to -54°C (-65°F) and the indicator properly connected but no power applied. After a period of 72 hours, the pressure in the chamber shall be reduced to the equivalent of 100,000 ±500 feet and power applied. The time required for the indicator to start operating shall not exceed 5 minutes. There shall be no binding or sticking at any indication. After the pressure has increased to atmospheric and the indicator has returned to room temperature, it shall meet all individual tests.

4.6.26 Temperatue shock. The temperature shock test shall be conducted in accordance with MIL-STD-810C, Method 503.1, procedure I. At the conclusion of this test, the indicator shall meet all individual tests.

4.6.27 Fungus. The fungus test shall be conducted in accordance with MIL-STD-810C, Method 508.2, procedure I. There shall be no sticking or binding in the pitch trim knob or damage to the indicator. The tolerances specified for the individual tests shall apply.

4.6.28 Salt fog. The salt fog test shall be conducted in accordance with MIL-STD-810C, Method 509.1, procedure I. At the end of the 48-hour drying period, there shall be no sticking or binding in the pitch trim knob or damage to the indicator. The tolerances specified for the individual tests shall apply.

4.6.29 Vibration failure. The vibration failure test shall be conducted with the indicator mounted in its normal operating position under the conditions specified in 4.3.1 through 4.3.3. The indicator shall be vibrated with a circular motion of 0.018 to 0.020 inch diameter in a plane inclined 45° to the horizontal plane. The

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frequency shall be varied uniformly from 5 to 50 Hz and return to 5 Hz once each hour for a period of 3 hours. At the conclusion of this test, the indicator shall meet the tests specified in 4.6.3 and 4.6.6. No looseness in the mechanism nor damge to any part of the indicator shall result from this test.

4.6.30 Acceleration (without power applied). The indicator, not operating, shall be mounted in its normal operating position and accelerated first along its vertical axis and then along each of two axes that are perpendicular to the vertical axis and to each other. I shall be subjected to an acceleration of 20g's in each of the above axes for a period of 1 minute. At the end of this acceleration, the indicator shall meet the individual tests. No damage to the indicator shall result from this test.

4.6.31 Pressurization. The indicator shall be placed in a chamber at an absolute pressure of 57 inches Hg and a temperature of 25°C (77°F) for a period of 4 hours. During this period, no deformation or other damage to the indicator shall occur. At the conclusion of this test, the indicator shall be closely examined for evidence of deformation and shall operate satisfactorily. The indicator shall then be subjected to and shall pass the test specified in 4.6.10. Evidence of deformation, damage, or any unsatisfactory operation shall be cause for rejection.

4.6.32 Electromagnetic interference. The electromagnetic interference test shall be conducted as required by MIL-STD-461 for Class 1B equipment and as described by MIL-STD-462.

4.6.33 Realiabililty and MTBF. The indicators shall be properly connected and subjected to a cycle consisting of a 4-hour perod of operation followed by a period of 5 minutes with no power applied to the indicator. During the operational periods, both axes of the attitude display shall be oscillated through a minimum of 30° at a rate of 5 to 7 times per minute, the lighting circuit shall be energized with 5V ac or dc, and the power warning flag system shall show that the indicator is operating. During the resting or cooling-off periods, all power shall be removed from the instrument under test, including power to the lighting and flag circuits, the standard test transmitter shall remain excited, and its rotor shall continue to be displaced through a minimum or $\pm 30^{\circ}$ at a rate of 5 to 7 times per minute. At least once during each 4-hour period, the pitch trim knobs shall be rotated to their extremes 20 times in each direction. During this test, the individual tests shall be conducted every 250 hours. The indicators shall be subjected to a reliability test in accordance with MIL-STD-781, test plan III, except each of the three indicators shall be operated for a minimum of 1,500 hours for a total of 4,500 which shall demonstrate an MTBF of 1,500 hours. At the option of the contractor, this test may be conducted on five indicators for a minimum of 900 hours of operation for each indicator. In the event of a reject decision, all shipments shall be discontinued until an evaluation has been made, corrective action taken, and the new item has been resubjected to this test. All out-of-tolerance condition shall be as follows:

a. Where a tolerance-type requirement is specified, a failure shall be considered as any electrical or mechanical condition which results in an increase of 100 percent or more of the specified tolerance.

b. Where a maximum-or minimum-type requirement is specified, a failure shall be considered as any electrical or mechanical condition which results in a value which is greater than the specified maximum or less than the specified minimum by more than 5 percent.

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c. Where a condition is specified as a requirment, a failure shall be considered as any physical, electrical, or mechanical condition which fails to comply with the requirements of this specificiation.

4.6.34 Longevity. One indicator, having passed the reliability test, shall continue to be tested through a longevity test for a total of 2,000 operating hours unless this time has been exceeded during the reliability test. Parts requiring replacement due to wearout within this period of time and the normal operating period of such parts shall be reported by the contractor to the contracting activity. At the end of this test, the pitch trim torque shall be between 2 and 6 ounce-inches.

4.7 Inspection and test. Tests for methods of preservation and packaging will be accomplished in accordance with Section 4 of MIL-P-116 to assure compliance with Section 4 of this specification.

5. Packaging

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Indications shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Indicators shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation Application. Preservative shall not be used.

5.1.1.4 Unit Packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method IIc of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of indicators shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicators packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicators packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless

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otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicators shall be packed in accordance with ASTM D3951.

5.3 Marking. In adddition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. Notes

6.1 Intended use. The ARU-13/A and ARU-20/A remote attitude indicators covered by this specification are intended for use in high performance aircraft. The indicators are controlled by a vertical reference control which provides synchro pitch and roll output signals.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Type of indicator required (see 1.2)

d. When sampling plan B tests will not be conducted

e. Levels of packaging and packing required (see section 5).

6.3 Data. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) referencing the appropriate data item description in the military departments' Authorized Data List (ADL).

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List No. 27710 whether or not such products have actually been so listed by that date. The attention of contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the OC-ALC/MMEOR, Tinker AFB, OK 73145-5990, and information pertaining to qualification of products may be obtained from that activity.

6.5 Subject term (key word) listing.

Attitude
Indicator
Remote

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6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Preparing Activity:
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PROJECT NO.
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-27701D	Indicator, Attitude, Remote

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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